Robotics and Autonomous Systems 199 (2026) 105360

Contents lists available at ScienceDirect

Robotics and Autonomous Systems

journal homepage: www.elsevier.com/locate/robot

Human–human and human–robot co–manipulation: A biomechanical analysis of a joint carrying task

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ARTICLE INFO

Keywords:
Assistive robot
Biomechanics
Human–robot collaboration
Human–robot interaction
Joint angles
Joint carrying
Spatio–temporal parameter

ABSTRACT

Assistive robots can support collaborative manipulation tasks such as carrying heavy or extended objects. As the human–human interaction is the basis for human–robot interaction, it is important to understand and quantify primarily haptic interaction. The subjects' movements were recorded with a 3D motion capture system to determine spatio–temporal and upper and lower body kinematic parameters. The human–human interaction provided foundational data on human movement in collaborative manipulation tasks. The task with the robot revealed almost no changes in upper body kinematics, however, it was slower and showed adaptations of the human movement in the center of mass motion and in spatio–temporal parameters and lower body kinematics. This shows, that analyzing the interaction between humans and assistive robots focusing on human movement is essential for further developing assistive robots.

1. Introduction

The loss of independence caused by reduced mobility and muscle strength [1,2] is one of the major problems our aging society faces [3,4]. Besides adapting the domestic environment [5], assistive robots have already adopted assistive tasks, such as cleaning and handing over objects [6,7]. Joint handling objects in a collaborative interaction with assistive robots may further contribute to regain or maintain independence of elderly individuals.

Collaborative interaction tasks can be broadly categorized according to the degree of coupling between partners. This ranges from coexistence in a shared workspace, to sequential collaboration in which agents contribute to the same goal or object by taking turns, to simultaneous collaboration in which agents work together in time and space, but without tight interdependent control, and ultimately to supportive collaboration, in which agents jointly manipulate the same object at the same time toward a shared goal [8,9]. Representative supportive collaboration tasks include joint carrying furniture, transporting or jointly moving extended or heavy objects, all of which require continuous mutual adaptation and object–mediated communication [10,11]. Therefore, joint handling extended or heavy objects in a collaborative interaction falls into the category of 'supportive collaboration' within the taxonomy of McGirr et al. [9], which is characterized by high interdependence, close physical proximity and sustained co–manipulation. Within the framework of Onnasch and Roesler [8], a supportive collaboration task with assistive robots in a leader–follower dyad can be further classified as a collocated and synchronous collaboration with low to moderate robot autonomy and an asymmetric leader–follower role distribution, in which the human initiates and guides the motion and the robot provides assistive support [8,12]. From a human–human collaboration perspective, the task is a form of joint action. It requires coordination in space and time in order to achieve a shared goal, and relies on mutual prediction and shared task representations [10,13] and can depend strongly on implicit coordination mechanisms, where forces and object dynamics act as communication channels between human partners [11,14,15].

Human–robot collaborative manipulation requires the integration of the human and its behavior into the control loop of the robotic system [16]. Understanding and quantifying human behavior in such

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https://doi.org/10.1016/j.robot.2026.105360
Received 23 April 2025; Received in revised form 29 December 2025; Accepted 19 January 2026
Available online 24 January 2026
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tasks could provide a knowledge base and metrics for the integration of the human into the control loop [17,18]. Respective control parameters obtained through the interaction are limited to the robot's sensors, and have already been successfully used for joint carrying tasks that involved human–robot dyads using haptic–only [17,19–23], haptic and visual [24–27], and visual–only [17,28] sensors. Mielke et al. [17] reported a significant decrease in walking velocity in a human–robot dyad with primarily visual interaction compared to an unconstrained human–human interaction in one–dimensional joint–carrying tasks of an extended object. The velocity in the primarily visual human–robot interaction was comparable to human–human dyads using primarily haptic interaction [17]. Similar results were found by Jensen et al. [18] for joint carrying a table in a human–human dyad in which the lead human walked backward; using primarily haptic interaction, a slower walking velocity and a longer duration of the task was observed compared to an unconstrained interaction [18].

In human–human interaction, verbal communication is not necessary [29,30]. Instead, motor communication strategies are often used to communicate [31,32], which can lead to synchronization effects and adaptations to movement and dyad kinematics [32,33]. Compared to solo walking, the walking velocity and step length is reduced and the ground contact time is increased in human–human collaborative carrying tasks [33]. The overall Center of Mass (COM) vertical displacement on the other hand, does not change [33].

Furthermore, the horizontal COM trajectories in human–human collaborative manipulation tasks were studied by Maroger et al. [34,35]. In human–human dyads carrying a table with a variety of goal positions and configurations of whether only one or both subjects knew the goal positions, the subjects COM and the table COM trajectories were analyzed. A shared strategy could not be observed between pairs of subjects, but showed that the trajectories taken were subject to high variability [34,35]. The attempt to predict the COM trajectories of the subjects by simulation was difficult to impossible presumably due to unpredictive human behavior and non-verbal communication [34,35]. Nevertheless, a further study by Maroger et al. showed that predicting the trajectory of the collaboratively carried object seems possible and may help to proactively improve a simulated human–robot interaction [36]. In all recorded trials of Maroger et al. [36] both subjects positioned themselves with the table in front of them, causing one subject to walk forward and one to walk backward.

In general, when a dyad collaboratively carries an object, the leading person either walks forward or backward. Schuengel et al. [23] analyzed a collaborative manipulation task to carry a table while walking forward in human–human and human–robot dyads. Their results suggest an adaptation of human movement to the task performed with the robot. Especially affected were the lower body kinematics, COM vertical displacement and spatio–temporal parameters. Whether the adaptations were due to a slower overall velocity in the human–robot dyad or constraints set by the robot remained open to future research [23]. Although forward walking while leading the dyad seems to be the preferable option with a symmetrical perceptual access, a backward walking condition would represent a coordination regime with asymmetric perceptual access. This has been shown to increase coordination demands and emphasis leader–follower dynamics in joint action and physical collaboration. [11,37].

Generally in backward walking, reducing the walking velocity leads to shorter step lengths [38–40] and to a reduced step frequency [39,41]. The effects of those changes in velocity, step length and frequency in backward walking on peak angular displacements and the angular Range of Motion (ROM) are evident [41]. These effects are also present in forward walking [42], which can help to understand changes in backward walking mechanics, since sagittal joint angle kinematics in backward walking can be considered time reversed of that of forward walking [38,43–45].

It is desirable to know the spatio–temporal parameters and kinematics of such a collaborative task within the specific class of supportive collaborative transport tasks with increased asymmetrical coordination demands due to backward walking and leading, carried out by two humans as a model for collaborative human–robot manipulation.

Therefore, the objectives of this study are to describe upper and lower body kinematics and spatio–temporal gait parameters in a collaborative carrying task with another human and a robot in order to obtain gait parameters in human carrying dyads and describe possible differences between the human–human and human–robot dyads. It is proposed that upper body kinematics are similar interacting with a robotic partner compared to a human partner. In addition, a slower backward walking velocity in the human–robot interaction than in the human–human interaction is hypothesized. Therefore, shorter step lengths and smaller step frequencies as well as a reduction of ROM in the sagittal plane joint angles of the lower extremities when paired with the robot were anticipated. In a constrained experimental area, a motion capture system was used to record the motion of the participants, as they collaboratively carried a table, first interacting primarily freely with a human partner and then with a hand–guided robot that relied mainly on haptic interaction. The participants walked backward while holding the table in front of them. The participants' spatio–temporal gait parameters and joint angle kinematics were analyzed and evaluated. The task was a joint team effort within the specific class of supportive collaborative transport tasks, the interaction between the human and the robot is described as collaborative [9].

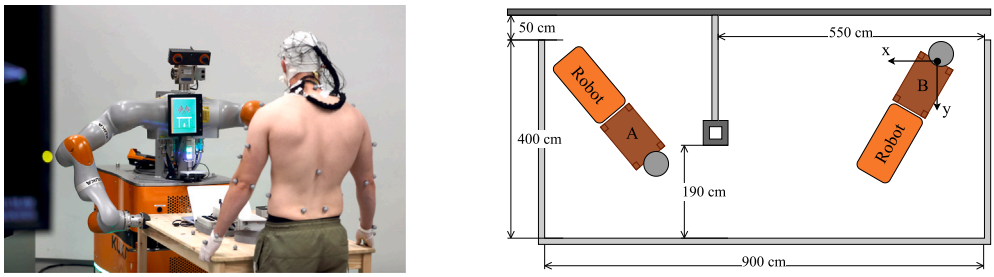
2. Methods

2.1. Subjects

Seven male and seven female volunteers provided their informed written consent to participate in this study, which was approved by the ethics committee of the German Sport University, Cologne (№ 132/2020). The experiments were conducted at KUKA Deutschland GmbH, Augsburg, Germany. One of the recruited subjects did not complete the experiment. Data were analyzed for $n = 13$ subjects (age: 30.9 ± 9.6 years, body height: 174.8 ± 9.7 cm, body mass: 69.1 ± 13.1 kg, all M±SD).

2.2. Study design and experimental protocol

The task was to carry a table (L×W×H = 120 ×75 ×73 cm, 14 kg) with a human partner, specially provided by the study team, from position A to B through an opening of 190 cm width, before repeating the experiment again with the robot (Fig. 1(a) and Fig. 1(b)). The table's positions were marked on the floor to indicate the beginning and end of the task. The participants had the leading role and walked backwards holding the table in front of them (Fig. 1(a)) [23]. The interaction in the human–human dyad was unconstrained, while the interaction with the robot was primarily haptic, meaning that the robot was hand–guided [23,46,47]. The robot consisted of a three degree of freedom KUKA mobile platform equipped with two seven degree of freedom KUKA LBR iiwa robotic arms with a maximum payload of 14 kg each (Fig. 1(a)). The table was held by the two KUKA LBR iiwa with two passive grippers attached to the end–effectors equipped with a torque limiter protecting the system of exceeding its capacity. The lift height was set by the robot and identical for all subjects. The system responded to applied forces with respective translational movements and to applied moments with rotations in the respective rotational directions using online effective mass optimization as described in Ritter & Sharma [47]. The environment was monitored by a combination of a 3D–sensor, laser scanners and RADAR to prevent collisions of the overall carrying dyad with the physical limits of the test area or detected obstacles. Collision prevention was realized by virtually applying a counterforce or a counter moment depending on the distance to the area limits or an obstacle to reduce the system velocity; right up to a standstill [23,46]. A detailed description of the



(a) (b)

Fig. 1. (a) Subject collaboratively carrying the table with the assistive robot (copyright 2020 KUKA Deutschland GmbH, Augsburg, Germany). (b) Top view of the experimental area. The gray circle represents the subject; A and B indicate the two positions at the start and end of the task, respectively. The light gray lines indicate mobile partitions that limit the test area. The rectangle in the middle indicates a pillar and the thick dark gray line the laboratory wall. The global coordinate system is set on the ground at position B.

robot can be found in Schuengel et al. [23] and Braun et al. [46] and the underlying robot control algorithm is described in detail in Ritter & Sharma [47].

Six repetitions for each condition were performed. The first three repetitions of each condition were performed as a familiarization of the participants with the task and the carrying partners. Of the last three repetitions, the trial with the shortest time elapsed was analyzed, assuming to be the best performed trial. Participants answered questionnaires and interviews on trust during collaborative human–robot interaction, of which the results are published elsewhere [48].

2.3. Data collection

The subjects' movements were recorded using a three–dimensional motion capture system (Vicon Motion Systems Ltd, Oxford, UK), equipped with 15 infrared MX13 cameras operating at 100 Hz. For tracking, 56 retro–reflective spherical markers with a 19 mm diameter were attached to upper– and lower–body bony landmarks. Technical clusters consisting of three markers were attached laterally to the thigh, shank, and upper arm segments. A comprehensive list of all markers is provided in the Appendix in Table A.1 and Fig. A.1. Prior to dynamic data collection, a static reference was captured with the participants standing erect, feet parallel and hip width apart, defining the ankle, knee, and hip joints 0° reference positions; the arms were abducted 90° at the shoulder joints.

2.4. Data processing

The original 3D marker trajectories were automatically labeled with a predefined virtual model (Vicon Nexus 2.11, Vicon Motion Systems, Oxford, UK). Gaps were filled using the rigid body gap fill function. If not enough markers of a segment were visible, the gap was filled via the pattern fill function. Both fill functions use either three remaining markers of the same rigid body or one marker as pattern source to fill marker trajectory gaps, respectively. The labeled marker trajectories were subsequently filtered with a fourth–order, zero–lag low–pass Butterworth filter with a cutoff frequency of 10 Hz.

Touch Down (TD) and Foot Off (FO) events were manually estimated by visual inspection of the 3D marker trajectories by a trained observer. TD events were defined as the first instance after a swing phase in which the toe marker stopped moving in the vertical and horizontal directions. FO events were defined as the first instance in which the heel and toe markers started to move continuously in the vertical or horizontal direction after the stance phase. These criteria were previously proven reliable [49].

Task duration was the time elapsed between the first FO to the last TD. Step frequency was the number of steps divided by the task duration. Path length was the total horizontal distance covered by the COM during the task. Step length was the path length divided by the number of steps. The ground contact time was the elapsed time between the TD and the FO event of the same foot. The duty factor was calculated as the ratio of the ground contact times of the left and right foot within one gait cycle and the total gait cycle time [50]. Step width as the lateral distance between both heel markers was calculated with respect to the pelvic local coordinate system, as described below.

The overall subject's COM position was calculated according to Minetti et al. [51] using the body segment parameters from Dempster et al. [52]. The velocity was the time incremental derivation of the position of the COM. The COM and hand vertical displacements were defined as the differences in height to the initial ground contact in each gait cycle. Additionally, the COM–hand distance was determined between the COM and the midpoint between the right and left hand, to represent the distance of the table to the subject (Appendix Fig. A.1).

The local coordinate systems for the upper body segments (thorax, upper arm, and forearm) were defined in accordance with the ISB recommendations [53]; the local coordinate systems for the lower body segments (pelvis, thigh, shank, and foot) were defined based on Zatsiorsky [54] and Hamill et al. [55]. The hip joint centers were determined according to Harrington et al. [56] and the glenohumeral joint centers were determined according to Rab et al. [57]. Shoulder and elbow joint angles were calculated as described in the International Society of Biomechanics (ISB) recommendations [53]. Hip, knee and ankle joint angles were calculated according to Zatsiorsky [54] and Hamill et al. [55]. Additionally, the thorax angles were calculated with respect to the pelvis local coordinate system.

COM vertical displacement, hand vertical displacement, and COM–hand distance time courses were calculated for each gait cycle and were time normalized by the duration of the gait cycle. The COM and hand amplitudes as the differences between the maximum and minimum values of the vertical displacements and COM and hand heights as their average heights were calculated for each gait cycle. Ground contact times, duty factors, and step widths were also calculated for each gait cycle. Upper body joint angle time courses were calculated for each gait cycle and were time normalized by the duration of the gait cycle and lower body joint angle time courses were calculated for each stance phase and were time normalized by the duration of the stance phase. The respective ROMs as the differences between the maximum and minimum values of the joint angles and the average joint angles as the average values of the joint angles were calculated for each gait cycle or stance phase. All parameters were averaged per subject for each condition. In order to go beyond the analysis of the average parameter values and their magnitude, and to investigate whether possible differences between the conditions were transient within or sustained throughout the step or gait cycle, the parameters were examined using Statistical Parameter Mapping (SPM) analysis.

To reduce data, only the right side was analyzed for the joint angles of the lower and upper extremities and the spatio–temporal parameter where applicable.

Table 1
Spatio–temporal parameters and COM kinematics while carrying a table with another human or an assistive robot and their differences between the two conditions.

Parameter	H–H M (SD)	H–R M (SD)	Diff. M (SD)	95 % CI	Diff. in % M (SD)	Test statistics	p	Effect size
Velocity (m/s)	0.69 (0.11)	0.27 (0.07)	−0.42 (0.11)	−0.5 to −0.4	-61 (11)	t(12) = −13.48	<0.001	d = 3.74
Froude number	0.06 (0.02)	0.01 (0.01)	−0.05 (0.02)	−0.1 to 0	−84 (9)	t(12) = −9.94	<0.001	d = 2.76
Task duration (s)	10 (2)	31 (10)	21 (9)	15.8 to 27.2	225 (106)	t(12) = 8.20	<0.001	d = 2.27
Step width (mm)	165 (69) [a]	150 (61) [a]	−3 (31) [a]	−51.7 to 13.8 [b]	-1 (28)	W = 91 (+36,−55)	0.542	r = 0.21
Step length (mm)	445 (59)	239 (54)	−206 (78)	−253.2 to −158.8	-45 (13)	t(12) = −9.51	<0.001	d = 2.64
Step frequency (1/min)	96 (16)	69 (15)	−28 (12)	−35 to −20.3	-28 (12)	t(12) = −8.19	<0.001	d = 2.27
Ground contact time (s)	0.8 (0.17) [a]	1.39 (0.48) [a]	0.36 (0.52) [a]	0.3 to 0.7 [b]	56 (52)	W = 91 (+91,−0)	<0.001	r = 1
Duty factor	0.62 (0.02)	0.73 (0.04)	0.11 (0.04)	0.1 to 0.1	18 (7)	t(12) = 9.4	<0.001	d = 2.61
COM height (mm)	1023 (105) [a]	1023 (97) [a]	2 (4) [a]	−7.7 to 3.9 [b]	0 (0)	W = 91 (+55,−36)	0.542	r = 0.21
COM amplitude (mm)	24 (8)	14 (7)	−10 (7)	−14.3 to −6.4	-42 (18)	t(12) = −5.67	<0.001	d = 1.57
COM height (%BH)	58.56 (0.93) [a]	58.64 (1.21) [a]	0.09 (0.23) [a]	−0.5 to 0.2 [b]	0 (0)	W = 91 (+55,−36)	0.542	r = 0.21
COM amplitude (%BH)	1.38 (0.43)	0.8 (0.4)	−0.58 (0.35)	−0.8 to −0.4	-42 (18)	t(12) = −5.99	< 0.001	d = 1.66
Hand height (mm)	922 (53)	898 (26)	−24 (37)	−46 to −1.3	-2 (4)	t(12) = −2.31	0.039	d = 0.64
Hand amplitude (mm)	23 (8) [a]	12 (4) [a]	−9 (5) [a]	−14.2 to −6.4 [b]	-46 (15)	W = 91 (+1,−90)	<0.001	r = 0.98
Hand height (%BH)	52.78 (1.53)	51.5 (2.1)	−1.28 (2.07)	−2.5 to 0	-2 (4)	t(12) = −2.24	0.045	d = 0.62
Hand amplitude (%BH)	1.28 (0.53) [a]	0.74 (0.21) [a]	−0.55 (0.22) [a]	−0.8 to −0.4 [b]	-46 (15)	W = 91 (+1,−90)	< 0.001	r = 0.98
COM-hand distance (mm)	201 (32)	170 (30)	−31 (32)	−49.9 to −11.6	-15 (15)	t(12) = −3.49	0.004	d = 0.97

H–H: Human–human dyad, H–R: Human–robot dyad, M: Mean, SD: Standard deviation, COM: Center of mass, Diff.: Deviation from robot to human carrying partner, Diff. in %: Deviation from robot to human carrying partner as percentage of the value of H–H value, CI: Confidence interval, t: t–value,
p: p–value, %BH: percent of body height, Cohen's d: Absolute effect size, W: Sum of signed ranks (sum of positive, sum of negative ranks), r: Matched pairs rank–biserial correlation.

[a] Wilcoxon signed rank test applied and values are expressed as median (interquartile range).

[b] Equivalent of CI for non–parametric parameters.

2.5. Statistical analysis

Statistical analysis was performed with R (version 4.1.0). The assumption of normal distribution of the subjects' means for each parameter was assessed with the Shapiro–Wilk test with the level of statistical significance set to $\alpha = 0.05$. Accordingly, a two–tailed paired t–test or a two–tailed paired Wilcoxon signed–rank test was used to identify differences between conditions, both with the level of statistical significance set to $\alpha = 0.05$. Data are reported as the mean and standard deviation or median and interquartile range [58] per condition, respectively. Additionally, Cohen's d was determined as absolute effect size, defined as small (0.2), moderate (0.5) or large (0.8) [59]. As an equivalent to effect sizes, the matched pairs rank-biserial correlation was computed for not normally distributed data [60]. In order to identify differences in the parameters as functions of time between conditions, a SPM analysis was performed, which was assessed with a two–tailed paired t–test and performed with the spm1d toolbox (version M.0.4.7) for MATLAB (version 2020b) [61]. The level of statistical significance was set at $\alpha = 0.05$.

3. Results

3.1. Spatio-temporal parameters

Finishing the task with the assistive robot took significantly longer 21 ±9 s, $p < 0.001$ than with a human carrying partner and the backward walking velocity was significantly slower −0.42 ± 0.11 m/s, $p < 0.001$ (Table 1). The duty factor and the ground contact time were significantly higher with the robotic partner (both $ps < 0.001$) (Table 1). The step frequency and step length were significantly reduced with the robotic partner (both $ps < 0.001$) (Table 1). The step width showed no significant differences (Table 1). The average COM height showed no significant difference. Its amplitude was significantly smaller ($p < 0.001$) for the interaction with the robot (Table 1). The hand height was significantly lower ($p = 0.039$) with the robot, its amplitude was significantly smaller ($p < 0.001$) (Table 1). The average COM–hand distance was significantly shorter ($p = 0.004$) when the robot was the carrying partner (Table 1). The SPM analysis showed a significantly smaller COM vertical displacement with the robot in the early (1% to 10%), mid (23% to 47%) and late (69% to 95%) gait cycle (all $ps < 0.001$) (Fig. 2b). It also showed a significantly smaller hand vertical displacement for the human–robot interaction in the mid gait cycle (27% to 41%, $p < 0.001$) (Fig. 2d). The SPM analysis of the COM–hand distance revealed that it was significantly shorter with the robot throughout the entire gait cycle ($p = 0.003$) (Fig. 2f).

3.2. Kinematics of the upper body

Carrying a table with an assistive robot did not show significant differences in the average angles or ROMs of thorax flexion/extension and thorax rotation, nor in the average angles or ROMs of the shoulder joint flexion/extension and rotation, and elbow joint flexion/extension (Table 2). The average shoulder joint abduction angle was significantly smaller ($p = 0.048$) and its ROM was significantly higher ($p = 0.002$) when carrying the table with the robot (Table 2). SPM analysis did not reveal any significant differences in the time courses of the upper body joint angles throughout the gait cycle (Fig. 3).

3.3. Joint kinematics of the lower body

Average dorsiflexion angle in the ankle joint was significantly smaller ($p = 0.037$) in the human–robot dyad. Average knee joint angle showed no significant differences between both conditions, nor did the average hip joint angle. ROM was significantly smaller when carrying the table with an assistive robot for the ankle joint angle, the knee joint angle, and the hip joint angle (all $ps < 0.001$) (Table 2).

SPM analysis (Fig. 2a,c,e) showed that the ankle joint dorsiflexion was significantly smaller in the human–robot dyad from 2% to 40% ($p < 0.001$) in the stance phase. Knee joint flexion was significantly smaller with the robot than with the human partner in the early stance phase (0% to 18%, $p = 0.024$) and significantly larger in the late stance phase (86% to 100%, $p = 0.033$). Hip joint flexion was significantly larger in the early stance phase (0% to 16%, $p = 0.043$) and significantly smaller in the late stance phase (94% to 100%, $p = 0.049$) when carrying the table with the robot.

4. Discussion

One objective of the present study was to quantitatively describe the upper and lower body kinematics and spatio–temporal parameters during the collaborative manipulation task of carrying a table in a human–human dyad with both humans facing each other carrying the

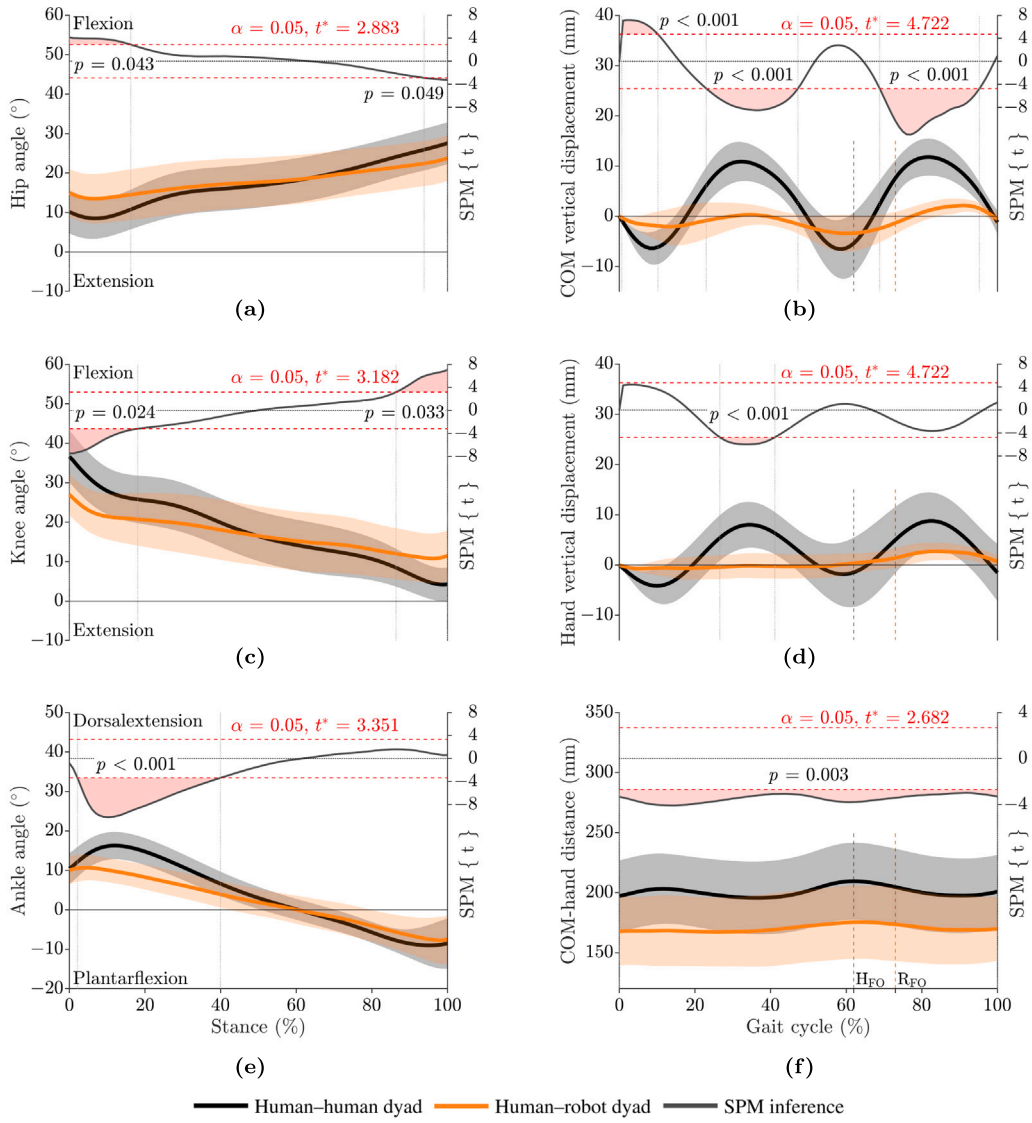


Fig. 2. The lower parts of the figures show, applied to the left y axes, the hip flexion/extension angle (a), knee flexion/extension angle (c), and ankle dorsiflexion/plantarflexion angle (e) as functions of the normalized stance phase and the COM vertical displacement (b), hand vertical displacement (d), and the COM–hand distance (f) as functions of the normalized gait cycle. The lines represent the respective averages, and the accordingly colored areas represent the standard deviation of carrying a table in the human–human dyads (black) or in the human–robot dyads (orange). H_{FO} and the black vertical dashed line indicates the mean FO with another human, R_{FO} and the orange vertical dashed line indicates the mean FO with the robot. The corresponding SPM inference is applied to the right y–axis at the top of the figures. The statistical significance level (α), critical thresholds (t^*), and, in the case of significant difference(s), the cluster–level probability value(s) (p) are shown. Ranges of significantly larger or smaller values are colored red in the SPM inference (gray line) and are framed by gray vertical dotted lines.

table in front of them. The subject analyzed walked backward and led the dyad. Our results agree with Schuengel et al. [23] who observed adaptations of human movement in a similar task when performed with a robot instead of a human.

In the human–human dyads, we observed an average backward walking velocity of 0.69 ± 0.11 m/s. Previous studies reported self–selected velocities in unconstrained backward walking between 0.94 m/s and 1.13 m/s [38,41,45,62–65]. Lanini et al. [33] found a reduction in velocity of 0.24 m/s between solo walking and carrying an extended object with another human; to mention in forward walking Sun et al. [64] reported a similar ground contact time of 0.79 s, compared to 0.80 s of the present study, at a backward walking velocity of 1.12 m/s in unconstrained backward walking. The average duty factor observed in the present study of 0.62 ± 0.02 describes walking, as it is higher than the threshold between walking and running of 0.5. Related values stated in the literature were found at velocities in unconstrained backward walking ranging from 0.87 m/s to 1.34 m/s [38,41,43,45,63,65,66]. Lee et al. [45] and Sun et al. [64] found similar step frequencies to that of the present study at velocities of 1.10 m/s and 1.08 m/s, respectively. Even though the tasks were different, it could be observed that the velocity in the human–human dyad was slower than the self–selected velocities in unconstrained backward walking reported in the literature [38,41,45,62–65].

When the task was performed in a human–robot dyad, the upper body kinematics were almost similar to these in the human–human dyad. The thorax angles remained statistically unchanged, while the shoulder joint abduction angle was, on average, 4 ± 7° smaller but with a 2 ± 2° greater ROM. However, these differences fall within the inter–subject standard deviation. The significantly shorter COM–hand distance (Fig. 2f) observed when the robot was the carrying partner showed that the table was carried closer to the hip. In contrast to the average COM height, the average COM amplitude was smaller in the

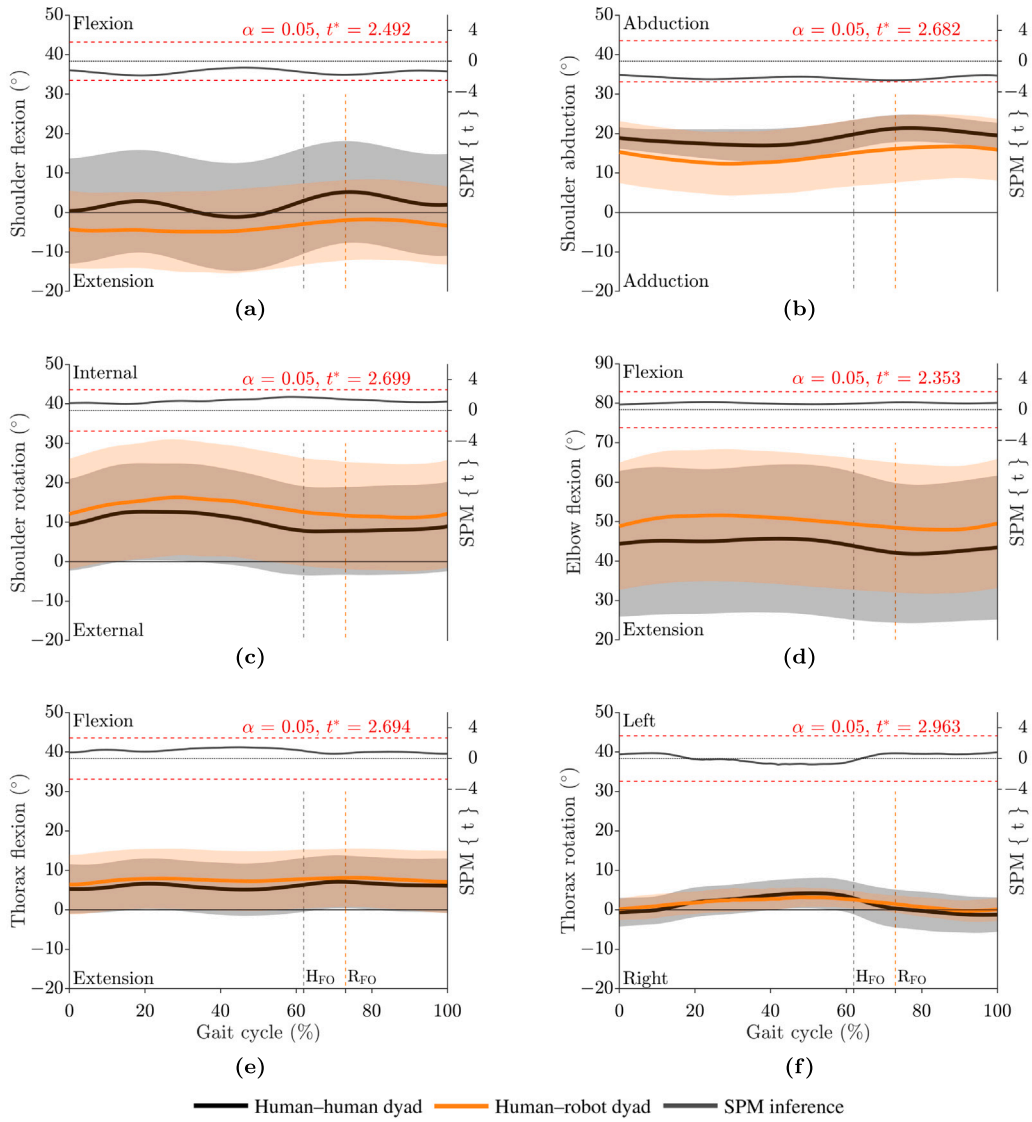


Fig. 3. The lower parts of the figures show, applied to the left y axes, the shoulder flexion/extension angle (a), shoulder abduction/adduction angle (b), shoulder internal/external rotation angle (c), elbow flexion/extension angle (d), thorax flexion/extension angle (e), and thorax left/right rotation angle (f) as functions of the normalized gait cycle. The lines represent the respective averages, and the accordingly colored areas represent the standard deviation of carrying a table in the human–human dyads (black) or in the human–robot dyads (orange). H_{FO} and the black vertical dashed line indicates the mean FO with another human, R_{FO} and the orange vertical dashed line indicates the mean FO with the robot. The corresponding SPM inference is applied to the right y–axis at the top of the figures. The statistical significance level (α), critical thresholds (t^*), and, in the case of significant difference(s), the cluster–level probability value(s) (p) are shown. Ranges of significantly larger or smaller values are colored red in the SPM inference (gray line) and are framed by gray vertical dotted lines.

human–robot dyad, as were both the hand height and its amplitude. The SPM analysis revealed that COM and hand vertical displacements oscillate with a magnitude of 24 ± 8 mm and 23 ± 8 mm, respectively, when performing the task in a human–human dyad. The COM vertical displacement is smaller than reported 39.3 ± 10.8 mm for unconstrained backward walking at a higher velocity of 0.99 m/s [67], but within the range of 20 mm to 46.6 mm, reported for forward walking at similar velocities of 0.6 m/s to 0.7 m/s [68–71]. The reduction in COM amplitude of 10 ± 7 mm with the robot partner could therefore be an effect of the slower velocity [67–70]. In addition, it could be affected by the fixed height of the table by the robot.

The spatio–temporal parameters and lower body kinematics showed greater differences than the upper body kinematics when the carrying partner was the robot instead of another human. When the task was performed in a human–robot dyad, the backward walking velocity was significantly slower, as expected, aligning with findings reported in the literature [17]. Our results further showed shorter step length, longer ground contact time, smaller step frequency and higher duty factor in the human–robot dyad. Previous studies that examined unconstrained backward walking reported shorter step lengths [38–40], longer ground contact times [41], smaller step frequencies [39,41], and a higher duty factor [41,63] with a reduction in backward walking velocity. The ROM in all the lower body joints was reduced in human–robot dyads. Results of Vilensky et al. [41] showed a reduction in lower body joint ROMs when the velocity was reduced and subsequently the step length was shorter [41]. The average ankle joint dorsiflexion was smaller with the robot, but only by 2° (Table 2). We observed a significantly reduced peak ankle dorsiflexion in the first half of the stance phase (Fig. 2(e)), a significantly reduced peak hip flexion in the second half of the stance phase (Fig. 2(a)) and a less extended hip in the first half of the stance phase (Fig. 2(a)) in the human–robot dyads compared to the human–human dyads. Since backward walking can be considered time–reversed forward walking regarding joint kinematics [38,43–45], the time–reversed results of Lim et al. [72] showed similar changes in lower body joint kinematics for velocity reduction in forward walking.

Table 2
Postural parameters while carrying a table with another human or an assistive robot and their differences between the two conditions. M and SD values are given in degree. Results highlighted in bold show significant differences.

Joint Angle		H–H M (SD)	H–R M (SD)	Diff. M (SD)	95 % CI	Diff. in % M (SD)	Test statistics	p	Effect size
Ankle joint dorsiflexion angle	**Average**	**4 (3)**	**2 (3)**	**−2 (3)**	**−3.2 to −0.1**	**−160 (221)**	**t(12) = −2.35**	**0.037**	**d = 0.65**
	ROM	**28 (3)**	**21 (5)**	**−7 (6)**	**−10.8 to −3.8**	**−25 (19)**	**t(12) = −4.57**	**< 0.001**	**d = 1.27**
Knee joint flexion angle	Average	18 (6)	17 (7)	−1 (5)	−3.6 to 1.9	−4 (33)	t(12) = −0.7	0.499	d = 0.19
	ROM	**33 (4)**	**19 (4)**	**−14 (5)**	**−16.6 to −10.5**	**−41 (12)**	**t(12) = −9.71**	**< 0.001**	**d = 2.69**
Hip joint flexion angle	Average	17 (6)	18 (6)	1 (4)	−2 to 3.3	5 (26)	t(12) = 0.54	0.6	d = 0.15
	ROM	**19 (3)**	**12 (3)**	**−8 (3)**	**−9.7 to −5.6**	**−39 (15)**	**t(12) = −8.07**	**< 0.001**	**d = 2.24**
Thorax Flexion angle	Average	6 (7)	8 (8)	1 (5)	−1.7 to 4.7	18 (144)	t(12) = 1.03	0.325	d = 0.28
	ROM	4 (1)	4 (1)	0 (1)	−0.5 to 1.3	15 (39)	t(12) = 1.04	0.317	d = 0.29
Thorax Rotation angle	Average	1 (4)	2 (3)	0 (5)	−3 to 3.4	−114 (632)	t(12) = 0.14	0.891	d = 0.04
	ROM	9 (3)	8 (2)	−1 (2)	−2.4 to 0.4	−6 (23)	t(12) = −1.53	0.153	d = 0.42
Shoulder joint flexion	Average	2 (14)	−4 (10)	−6 (14)	−14.3 to 3.1	−53 (226)	t(12) = −1.39	0.189	d = 0.39
	ROM	9 (3)	9 (4)	0 (3)	−2.3 to 1.4	−3 (31)	t(12) = −0.51	0.621	d = 0.14
Shoulder joint abduction	**Average**	**19 (4)**	**15 (8)**	**−4 (7)**	**−8.7 to 0**	**−24 (36)**	**t(12) = −2.2**	**0.048**	**d = 0.61**
	ROM	**7 (2)**	**9 (3)**	**2 (2)**	**1 to 3.4**	**33 (33)**	**t(12) = 3.88**	**0.002**	**d = 1.08**
Shoulder joint internal rotation	Average	10 (12)	14 (15)	4 (10)	−2.6 to 9.8	13 (145)	t(12) = 1.27	0.229	d = 0.35
	ROM	9 (3)	10 (4)	1 (3)	−0.7 to 2.9	17 (43)	t(12) = 1.35	0.202	d = 0.37
Elbow joint flexion angle	Average	44 (19)	50 (17)	6 (24)	−8.9 to 20.4	40 (92)	t(12) = 0.85	0.411	d = 0.24
	ROM	6 (3)	8 (4)	2 (4)	−0.4 to 4.1	49 (80)	t(12) = 1.78	0.1	d = 0.49

The given angle directions describe positive angle values; negative angle values indicate the opposing angular direction. H–H: Human–human dyad, H–R: Human–robot dyad, M: Mean, SD: Standard deviation, ROM: Range of motion, Diff.: Deviation from robot to human carrying partner, Diff. in %: Deviation from robot to human carrying partner as percentage of the value of H–H value, CI: Confidence interval, t: t–value, p: p–value, Cohen's d: Absolute effect size.

To the authors' knowledge, the present study has provided the first published set of human movement parameters associated with backward walking and holding a table in front of the body, leading a supportive collaborative manipulation task with another human with unconstrained interaction. Furthermore, the same parameters were also recorded when the same task was performed with a robot partner by using primarily haptic interaction. The results reported in this study were interpreted as characterizing interaction dynamics within the specific class of supportive collaborative transport tasks, rather than as generalizable to all forms of human–robot or human–human collaboration [8]. Particularly, the spatio–temporal and kinematic joint parameter data collected when the collaborative manipulation task was performed by human–human dyads provide a foundation for the development of robots designed for collaborative manipulation tasks. Furthermore, due to the slower velocity in human–robot collaboration, it can be assumed that the robot indirectly imposed a velocity constraint through the interaction mechanisms, in terms of how quickly and directly it followed human input. Therefore, the human potentially shared the leading role with the robot to a certain extent, in terms of velocity.

The human is able to perform the backward walking collaborative manipulation task with the robot by adapting his/her movement and posture. Schuengel et al. [23] reached the same conclusion when the task was performed walking forward. Assuming that to improve human–robot interaction is in the first step equivalent to maintaining the movement and posture of the human–human interaction of the same task, future development of interactive robotic systems should adopt human walking velocity, adopt table height to account for the human's anthropometric and consider human COM oscillation, thereby reducing corrective adaptations in human movement. Furthermore, interactive robotic systems designed for the specific class of supportive collaborative transport tasks should support human walking velocity to be truly self–selected and therefore allow the dyad to be completely human–led; while collision avoidance must always be enabled. Thus, appropriate parameters for optimization could be the translational and rotational accelerations of the robot. Since it is known that humans may adapt movements during collaborative carrying tasks [32,33], a further step could be to go beyond replicating human–human interaction and enable the robot to adapt to the individual movement and in favor of the postural load; even distributing more load to the robot using lifting torques. Another approach could be to detect human gait phases and joint configuration in real–time and adjust robot motion planning or compliance to preserve natural movement. Thus, investigating the biomechanics of movement during collaborative manipulation tasks in human–human dyads, as well as in human–robot dyads may be a valuable contribution to future interactive robotic development.

4.1. Limitations

Certain aspects regarding the interpretation of the results presented in the human–human and human–robot interaction require further consideration. Since walking velocities are known to affect COM oscillation, joint kinematics and spatio–temporal parameters, it is not possible to clearly distinguish whether the underlying cause for the observed differences is the slower velocity, the limited human–robot interaction or the constraints of the robotic system itself. Furthermore, the conditions for the human–human and human–robot interaction were not entirely equal, such as humans have multiple ways of interacting (visual, verbal, and haptic communication), while the human–robot interaction was primarily based on haptic information. Moreover, no information about the interaction forces was available, and therefore the load sharing between carrying partners and a possible influence of the interaction forces on kinematics and spatio–temporal parameters could not be determined. Based on the present study, further research should be conducted to better understand the possible relationships between ways of interaction or their limitations, movement velocity, interaction forces, and human movement parameters. The participants in this study were young and middle–aged to ensure that the study's goals were safely achieved. However, the robots are being developed to assist elderly people, and clearly our subjects did not represent the targeted users. Hence, future studies should include participants who better represent the target population.

4.2. Conclusion

In conclusion, our results contribute data on human movement in collaborative manipulation tasks. The findings further highlight that studies on the interaction between humans and assistive robots with the focus on the human movement are essential for further development of assistive robots. Additionally, the results combined with future research may help to better understand the underlying causes for the presented changes in human movement when interacting with the robot compared to a human interaction partner and thus, support their further development. Although the collaborative manipulation

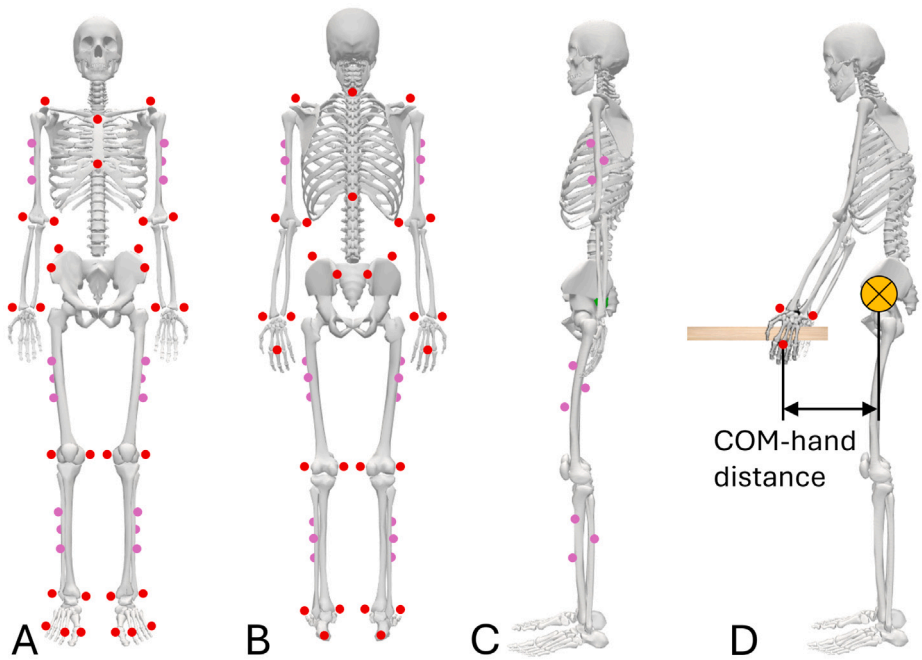


Fig. A.1. Comprehensive representation of all markers used in the current study at anatomical landmarks (red) and in clusters (purple) attached bilaterally to the limbs in the front view (A), back view (B), and in the side view (C, only clusters) and an unilateral schematic of a subject carrying the table with representation of the COM–hand distance (D) between the subjects overall COM (yellow circle) and the midpoint of both hand markers at the left and right mid metacarpale III.

was primarily analyzed to support development of assistive robots for domestic applications, it would also be possible to extend the targeted application area to industry or services, since collaborative carrying tasks are universal tasks.

CRediT authorship contribution statement

Fabian Goell: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Bjoern Braunstein:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Jule Heieis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis. **Daniel Braun:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Nadine Reißner:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Kirill Safronov:** Writing – review & editing, Resources, Project administration. **Christian Weiser:** Writing – review & editing, Resources, Investigation. **Verena Schuengel:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Kirsten Albracht:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Funding

This work was supported by the Federal Ministry of Education and Research as project 'Lernende roboter assistierte Systeme für das neuromuskuläre Training – RoSylerNT' (16SV7853/16SV7851) as part of the funding program 'Forschungsprogramm zur Mensch–Technik–Interaktion: Technik zum Menschen bringen'.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kirsten Albracht reports financial support was provided by German Federal Ministry of Education and Research. Daniel Braun reports financial support was provided by German Federal Ministry of Education and Research. Daniel Braun reports a relationship with KUKA Deutschland GmbH that includes: employment. Nadine Reissner reports a relationship with KUKA Deutschland GmbH that includes: employment. Kirill Safronov reports a relationship with KUKA Deutschland GmbH that includes: employment. Christian Weiser reports a relationship with KUKA Deutschland GmbH that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to thank all subjects who participated in the study and the project partners of KUKA Deutschland GmbH, Germany.

Appendix

Table A.1 and Fig. A.1.

Data availability

The data generated during the current study is available from the corresponding author upon reasonable request.

Table A.1
Comprehensive list of all markers used in the current study at anatomical landmark and in clusters on the extremity segments. Markers attached to the extremities were placed bilaterally.

Segment	Anatomical landmark
Foot	Metatarsal head II
	Side of metatarsal head I
	Side of metatarsal head V
	Heel
Shank	Medial malleolus
	Lateral malleolus
	Three–point cluster (dorsolateral)
Thigh	Medial epicondyle
	Lateral epicondyle
	Three–point cluster (dorsolateral)
Pelvis	Anterior superior iliac spine
	Posterior superior iliac spine
	Most proximal iliac crest
Torso	12th thoracic vertebra
	7th cervical vertebra
	Incisura jugolaris
	Processus Xyphoideus
	Acromion
Upper arm	Lateral epicondyle
	Medial epicondyle
	Three–point cluster (dorsolateral)
Forearm	Radial processus styloideus
	Ulnar processus styloideus
Hand	Mid metacarpale III

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